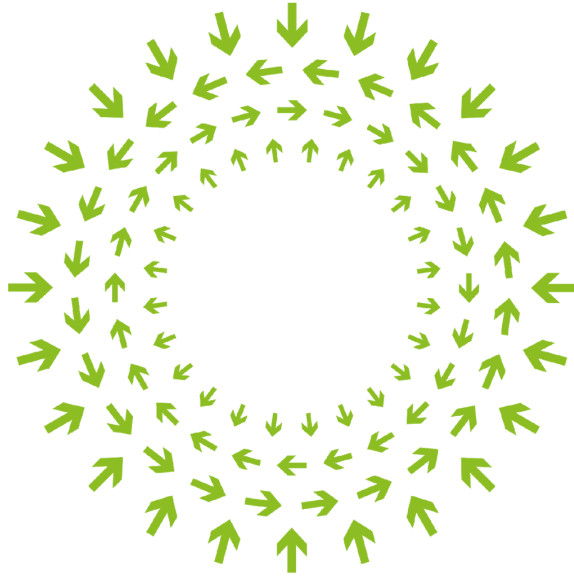




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Compute Project

Open Rack V3 HPR V2 Power Monitoring Module (PMM)

Rev: 1.0.0

08/04/2026

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Revision History

Date	Version	Change By	Comments
08/04/2026	1.0.0	Jingjing Sun Mohammed Alisa Ben Kim Wenying Zhang Yibing Tang Dmitriy Shapiro	Initial OCP Release

2. Scope

This design specification defines the technical requirements of the Power Monitoring Module (PMM) for the Open Rack V3 high power rack (ORV3 HPR) used in the Open Compute Project.

3. PMM Overview

The PMM is the power monitoring module for the Open Rack V3 HPR V2 platform. It is installed in each power shelf and battery shelf -- one PMM per shelf is required.

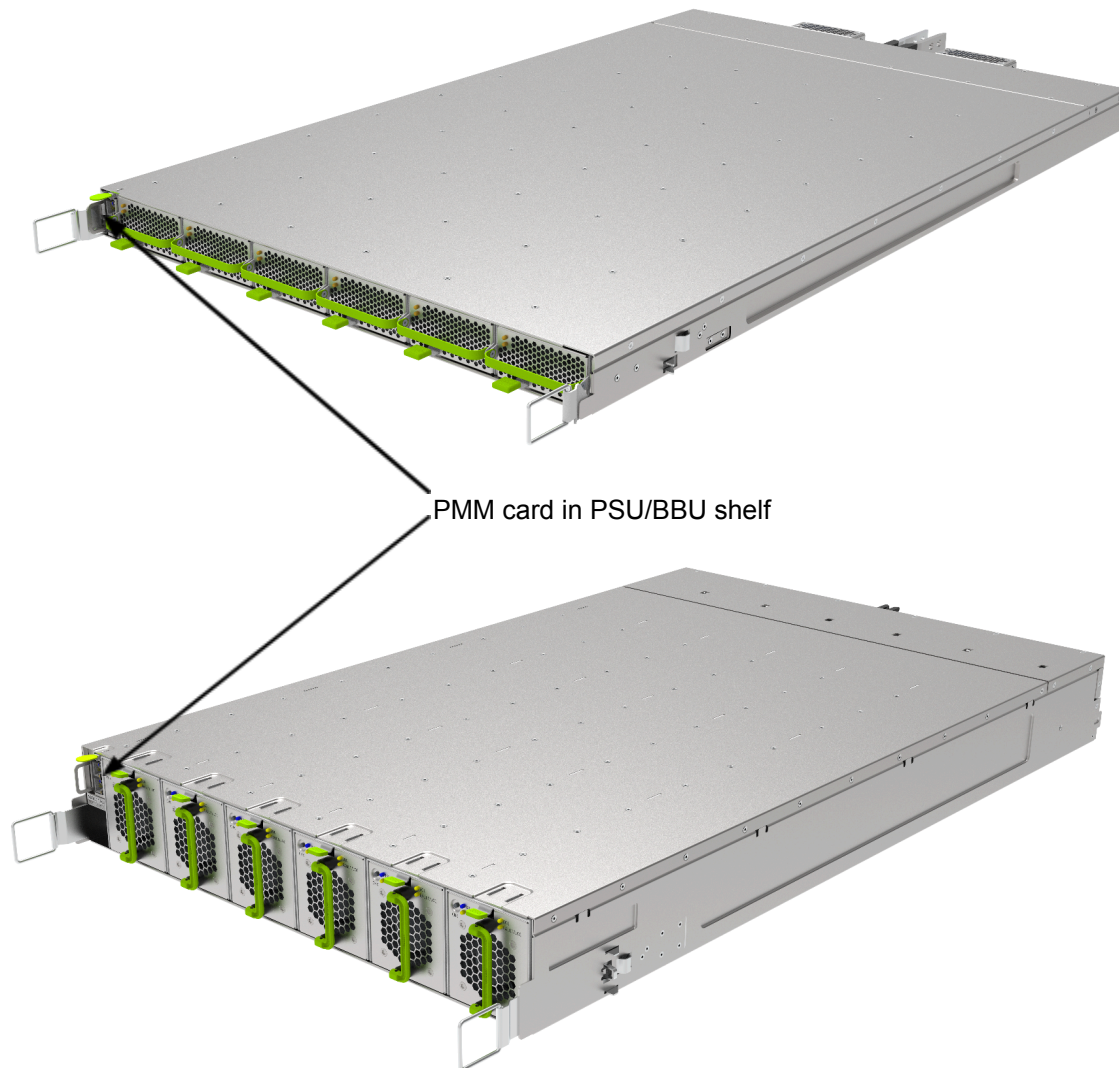


Figure 1: Isometric view of 72kW Power Shelf/BBU Shelf

The PMM card monitors the status of the PSU or BBU modules on the shelf. It provides Modbus service with aggregated module data to the upstream rack management device (Modbus master) via the RJ45 Modbus port.

The PMM design supports Modbus daisy chain connection to allow multiple shelves on the single Modbus daisy chain by automatically allocating each PSU or BBU shelf a unique Modbus address and CAN bus address if applicable.

Figure 2 shows a typical Modbus daisy chain connections and inter shelf control daisy chain connections on an ORV3-HPR rack with three PSU shelves, three BBU shelves and one rack management device.

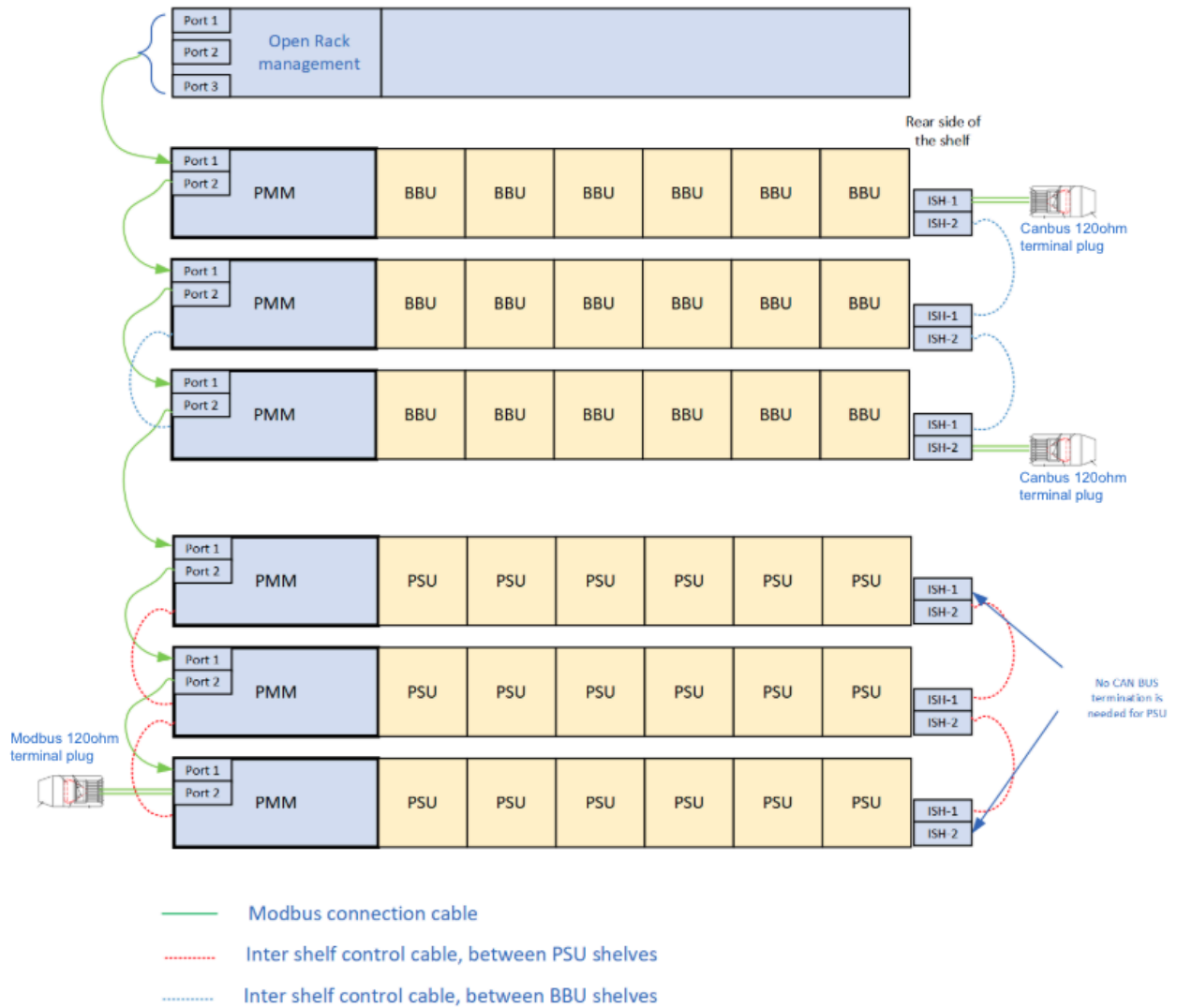


Figure 2: Typical Modbus daisy chain connections among ORV3-HPR rack with three PSU/BBU shelves and one rack management device

4. Electrical Requirements

4.1 Key Features of the Power Monitoring module (PMM)

- 1) ORV3 HPR continues using the RS485 Modbus as the communication interface between the rack management device and the PMM card through an 8-pin Ethernet cable. The PMM acts as a Modbus secondary.
- 2) The PMM supports multi-shelf daisy chain Modbus connection. The PMM supports the mix connection of the PSU and BBU shelves on the daisy chain.
- 3) The PMM automatically allocates and shares unique Modbus addresses and CAN bus addresses for the PMMs/Shelves and BBUs on the same daisy chain.
- 4) The PMM also communicates with PSU/BBU modules in the shelf via RS485 Modbus, through the PMM PCB edge card connector. Since the module uses the same communication interface as in ORV3, the changes on the PSU/BBU module register map and their firmware are minimized.
- 5) The PMM, with a microcontroller unit (MCU) included in the design, collects and consolidates the timing critical PSU or BBU parameters from all the modules within the shelf within a short time interval (~500ms to 1s) and share the information with the rack management device upon request.
- 6) The PMM collects all data from the modules within the shelf periodically (~10s) and provides Modbus read service to the rack management device.
- 7) The PMM provides the capability for the rack management device to indirectly access (read and write) the individual register on any module, with the sacrifice of read and write latency. The write access request from rack management shall be limited to configuration only. Usually, there is no need for the rack management device to read individual registers on the module.
- 8) The PMM provides the shelf level health information, such as chassis manufacturing information, PMM manufacturing information, Modbus communication status (CRC, timeout etc.), list of the available modules.
- 9) The PMM shall support online fail-safe firmware upgrade for both the PMM itself and the modules in the shelf. The firmware upgrade for PMM shall NOT cause any power interruption at rack level.
- 10) The PMM shall support hot swap of the PMM without affecting the key functionality of PSU and BBU shelves. Hot swap of the PMM shall NOT cause any power interruption at rack level.

Please note that the PMM is NOT designed to actively control the operation of the PSU or BBU module. The PSU and BBU module shall be able to operate on its own without any power interruption during the following scenarios:

- Firmware update and reboot.
- Failed the firmware upgrade and recovered to the previous firmware version.
- Reseat of the PMM card.
- Hot swap, replace and power up the PMM.

The rack management device is allowed to lose the communication to the shelves during the above scenarios, but communication shall recover after the above scenarios finish.

The BBU CAN bus communication may have some issues due to the conflict of CAN bus address if the Modbus cable was missing or not set up correctly, but it should not affect the transition between PSU and BBU.

4.2 High-Level Electrical diagram of PMM and Connectivity

Figure 3 below is the high-level electrical diagram for the PMM. Since the PMM is usually a part of the PSU or BBU shelf, the critical shelf and module signals related to the PMM are also included for reference.

Two RJ45 ports are available on the PMM front panel, designed for the Modbus communication and the inter shelf control. On the rear side of the PMM card is the PCB edge connector, providing electrical interface to the shelf and modules.

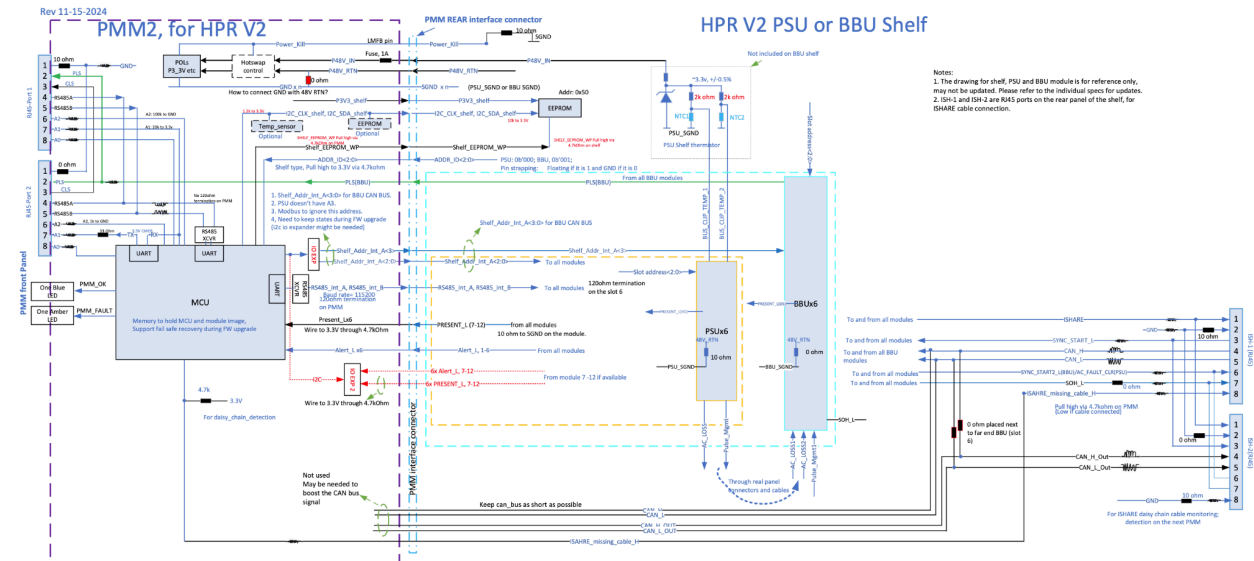


Figure 3: PMM high-level electrical diagram

Key differences from HPR V1 PMM,

1. two RJ45 ports for I2C cable are moved to the rear side of the shelf.
2. 6 alert and 6 present signals are added to PMM for future use. Suggest adding to I2C I/O expander to minimize the hardware changes from the HPR v1. This is a feature reserved for the future (firmware register map change is needed).

4.3 Electrical interface on the front panel

The PMM contains two RJ45 connectors located on the bulkhead of the assembly. The pinouts of the connectors are shown below. The RJ45 location shall be defined in the mechanical section. Looking from the front, top RJ45 is #1, bottom RJ45 is #2.

4.3.1 RJ45 #1 and #2

RJ45 port #1 and port #2 include the electrical connections for the Modbus communication, address pins and alert signals. Two port design allows daisy chain Modbus connection with multiple PSU shelves, BBU shelves and RPU on the same daisy chain.

RJ45 #1

Pin	Wire color	Function
1	White/Orange	GND
2	Orange	PLS (power loss siren, from BBU shelf)
3	White/Green	CLS (cooling loss siren, from the RPU cooling device)
4	Blue	RS485A
5	White/Blue	RS485B
6	Green	RS485_Addr2
7	White/Brown	RS485_Addr1
8	Brown	RS485_Addr0

Table 1: *RJ45#1 electrical connection pinout assignment**RJ45 #2*

Pin	Wire color	Function
1	White/Orange	GND
2	Orange	PLS (power loss siren, from BBU shelf)
3	White/Green	CLS (cooling loss siren, from the RPU cooling device)
4	Blue	RS485A
5	White/Blue	RS485B
6	Green	RS485_Addr2
7	White/Brown	RS485_Addr1
8	Brown	RS485_Addr0

Table 2: *RJ45#2 electrical connection pinout assignment*

Pin #1-7 and Pin #2-7 are used as the UART for serial communication between two PMMs. Specifically, the PMM will receive the data with pin #1-7 from the previous PMM and send data with pin #2-7 to the next PMM on the daisy chain. Refer to [5.1.1](#) for more details. Besides using as the UART, the Pin #1-8, #1-7, #1-6 shall be monitored by MCU to obtain the Modbus port address <pin6/7/8> from the rack management device.

The RJ45 #1 address pins 6,7 and 8 are connected either to the rack management device or previous PMM. For example, on the Wedge400 as the rack management device, the address pin has 10k ohm pull up to 3.3V for logic 1 and 33 ohm pull down to signal ground for logic 0. The pull-ups or pull-downs on the PMM design shall not affect the address settings.

4.3.2 Power loss siren (PLS)

Pin 2 of port #1 and Pin 2 of port #2 are for the BBU power loss siren (PLS) signal. On the PMM card, two pins are wired together and wire ORed with the PLS signals from the shelf. Through the Modbus daisy chain connection, the PLS signals from all shelves are joined together and monitored by the rack management device.

No pull up or pull down to the PLS on the PMM card. The signal is NOT monitored on the PMM card.

4.3.3 Cooling loss siren (CLS)

Pin 3 of #1 and pin 3 of #2 are reserved for the RPU cooling loss siren (CLS) signal. Since the signal is not related to the PSU and BBU module, two pins are wired together directly on the PMM. The RPU CLS signal will be monitored by the rack management device.

No pull up or pull down to the CLS on the PMM card. The signal is NOT monitored on the PMM card.

4.3.4 PMM RS485 Modbus interface

The RS485 Modbus is the only data communication interface between the rack management device and the PMMs on the power shelves. For simplicity, we will call the interface **the PMM Modbus** hereafter. The Modbus differential pair signals are connected to Pin4,5 of port #1 and port #2, and the MCU.

The Modbus design follows Modbus standard specifications [3], but may run at a higher baud rate.

There is another Modbus interface between the PMM and PSU/BBU modules. We will call it **the module Modbus** hereafter.

4.3.4.1 Modbus termination

Unlike the ORV3 which was designed to support up to two PSU shelves or two BBU shelves on a single Modbus daisy chain, the ORV3 HPR will support more than 3 shelves on the same Modbus port. In this case of having 8 PSU shelves on the same Modbus daisy chain, the load to the Modbus driver is too heavy if each shelf includes a 120-ohm termination resistor.

To solve the problem, ORV3 HPR will use external Modbus termination. One RJ45 termination plug is needed on the last open Modbus port (port #2 on the last PMM - either on PSU, BBU shelf or RPU).

Below is the termination plug drawing for reference. There is a 120-ohm resistor between pin 4 and pin 5. All those pins are floating.

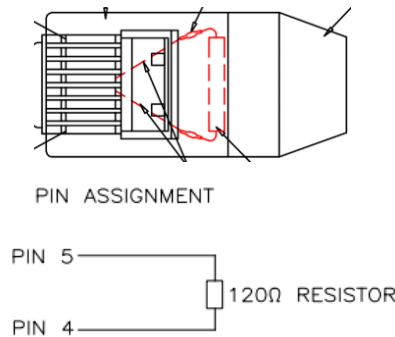


Figure 4: 120-ohm termination plug diagram.

4.3.5 Filtering

The following filters and ESD diodes are reserved to mitigate EMC and ESD, as shown in Figure 5. Components MPN are just for reference here.

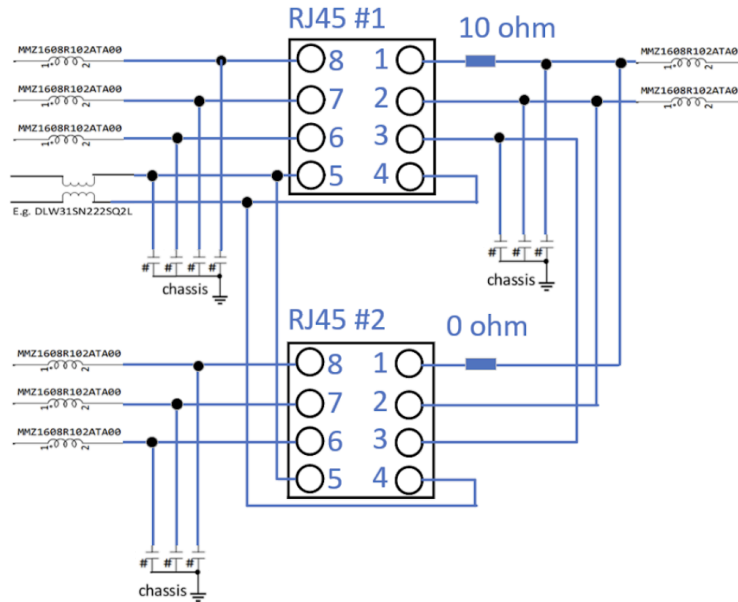


Figure 5: RJ45#1 and RJ45#2 pinout connection

4.4 Electrical interface to the shelf and module

The interface between the PMM card and the shelf/modules is an 84 pin PCB edge connector.

4.4.1 PMM Edge Connector

The signals across the PCB edge connectors can be categorized into the following groups:

- 48V Power supply and power_kill signal from shelf to the PMM
- 3.3V and I2c from the PMM to the shelf for housekeeping
- Shelf type ID from the shelf to the PMM
- Shelf EEPROM I2C interface and write protection
- Shelf addresses from the PMM to the modules for BBU CAN bus address control.
- Module Modbus between the PMM and the modules
- Module alert signals to the PMM
- Module present signals to the PMM
- Shelf level alarm signal, such as PLS
- Spare IO pins reserved for future growth.
- CAN bus signals routed to PMM but not used. Reserved for future if CAN bus booster is needed

Pin out of the edge connector is shown in table 3 below.

Signal Name	I/O (PMM)	Description	Pin # PMM2	Note HPR V2 PMM	PMM
ADDR_ID_0	I	Shelf ID	A1		4.7k to 3.3V

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ADDR_ID_1	I	000 1U power shelf	B1		4.7k to 3.3V
ADDR_ID_2	I	001 Battery shelf	A2		4.7k to 3.3V
GND	I	Ground	B2		GND to 48V RTN through 0 ohm
ALERT_0_N	I	PSU/BBU Alert	A3		
ALERT_1_N	I	PSU/BBU Alert	B3		
ALERT_2_N	I	PSU/BBU Alert	A4		
ALERT_3_N	I	PSU/BBU Alert	B4		
ALERT_4_N	I	PSU/BBU Alert	A5		
ALERT_5_N	I	PSU/BBU Alert	B5		
ALERT_6_N	I	PSU/BBU Alert	A6	Not used before	connect to i2C IO expander for future use
ALERT_7_N	I	PSU/BBU Alert	B6	Not used before	connect to i2C IO expander for future use
ALERT_8_N	I	PSU/BBU Alert	A7	Not used before	connect to i2C IO expander for future use
ALERT_9_N	I	PSU/BBU Alert	B7	Not used before	connect to i2C IO expander for future use
ALERT_10_N	I	PSU/BBU Alert	A8	Not used before	connect to i2C IO expander for future use
ALERT_11_N	I	PSU/BBU Alert	B8	Not used before	connect to i2C IO expander for future use
GND	I	Ground	A9		
RESET_0	O	PSU/BBU Reset	B9		Not used
RESET_1	O	PSU/BBU Reset	A10		Not used
RESET_2	O	PSU/BBU Reset	B10		Not used
RESET_3	O	PSU/BBU Reset	A11		Not used
RESET_4	O	PSU/BBU Reset	B11		Not used
RESET_5	O	PSU/BBU Reset	A12		Not used
RESET_6	O	PSU/BBU Reset	B12		Not used
RESET_7	O	PSU/BBU Reset	A13		Not used
RESET_8	O	PSU/BBU Reset	B13		Not used
RESET_9	O	PSU/BBU Reset	A14		Not used

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RESET_10	O	PSU/BBU Reset	B14		Not used
RESET_11	O	PSU/BBU Reset	A15		Not used
GND	I	Ground	B15		
NC (clearance)		No copper finger	A16	New	
PLS	I	PSU/BBU Power Loss Siren	B16	Changed pin number	
Shelf_Addr0	O	PSU/BBU Shelf MODBUS address	A17		Output of an IO expander, for CAN bus address
Shelf_Addr1	O	PSU/BBU Shelf MODBUS address	B17		Output of an IO expander, for CAN bus address
Shelf_Addr2	O	PSU/BBU Shelf MODBUS address	A18		Output of an IO expander, for CAN bus address
RS485A_Int	I/O	Shared PSU/BBU MODBUS (to PSU/BBU Modules)	B18		120 ohm termination on PMM
RS485B_Int	I/O	Shared PSU/BBU MODBUS (to PSU/BBU Modules)	A19		120 ohm termination on PMM
PRESENT_L_6	I	PRESENT_L for slot 6, pull high through 4.7kohm	B19	Added	4.7k to 3.3V, connect to i2C IO expander for future use
PRESENT_L_7	I	PRESENT_L for slot 7, pull high through 4.7kohm	A20	Added	4.7k to 3.3V, connect to i2C IO expander for future use
PRESENT_L_8	I	PRESENT_L for slot 8, pull high through 4.7kohm	B20	Added	4.7k to 3.3V, connect to i2C IO expander for future use
PRESENT_L_9	I	PRESENT_L for slot 9, pull high through 4.7kohm	A21	Added	4.7k to 3.3V, connect to i2C IO expander for future use
PRESENT_L_10	I	PRESENT_L for slot 10, pull high through 4.7kohm	B21	Added	4.7k to 3.3V, connect to i2C IO expander for future use
PRESENT_L_11	I	PRESENT_L for slot 11, pull high through 4.7kohm	A22	Added	4.7k to 3.3V, connect to i2C IO expander for future use
GPIO_SPARE1	I/O	GPIO input	B22	New on 12/19/2024	optional pull up to 3.3V and pull down to GND, do not install, ideally connect to IO expander, default high impedance
GPIO_SPARE2	I/O	GPIO input	A23	New on 12/19/2024	optional pull up to 3.3V and pull down to GND, do not install, ideally connect to IO expander, default high impedance
GPIO_SPARE3	I/O	spares, no need to connect	B23	New	Not used
GPIO_SPARE4	I/O	GPIO output	A24	New on 12/19/2024	ideally connect to IO expander
GPIO_SPARE5	I/O	GPIO Output	B24	New on 12/19/2024	ideally connect to IO expander

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GPIO_SPARE6	I/O	spares, no need to connect	A25	New	Not used
I2C_SDA_shelf	I/O	I2C for Shelf Temp and FRU	B25		1.2k to 3.3V
I2C_CLK_shelf	O	I2C for Shelf Temp and FRU	A26		1.2k to 3.3V
Shelf_Addr3	I/O	Add one bit to the PSU/BBU Shelf address, connect to BBU slot address A3.	B26		Output of an IO expander, for CAN bus address
PRESENT_L_0	I/O	PRESENT_L for slot 1, pull high through 4.7kohm	A27		4.7k to 3.3V
PRESENT_L_1	I/O	PRESENT_L for slot 2, pull high through 4.7kohm	B27		4.7k to 3.3V
PRESENT_L_2	I/O	PRESENT_L for slot 3, pull high through 4.7kohm	A28		4.7k to 3.3V
PRESENT_L_3	I/O	PRESENT_L for slot 4, pull high through 4.7kohm	B28		4.7k to 3.3V
PRESENT_L_4	I/O	PRESENT_L for slot 5, pull high through 4.7kohm	A29		4.7k to 3.3V
PRESENT_L_5	I/O	PRESENT_L for slot 6, pull high through 4.7kohm	B29		4.7k to 3.3V
CAN_H	I/O	reserved for future use	A30		Use external termination plug on the first shelf
CAN_L	I/O	reserved for future use	B30		Use external termination plug on the first shelf
NC (clearance)		No copper finger	A31		
GND	O	Ground	B31		
NC (clearance)		No copper finger	A32	New	
ISHARE_cable_missing_H	I/O	ISHARE cable missing detection	B32	New	4.7k to 3.3v
NC (clearance)			A33	New	
P3V3_shelf	O	3.3V for shelf I2C	B33		Power on PMM
NC (clearance)			A34	New	
GND	O	Ground	B34		
ADC1	A	spares	A35		Not used
ADC2	A	spares	B35		Not used
ADC3	A	spares	A36		Not used
ADC4	A	spares	B36		Not used

CAN_H_OUT	I/O	reserved for future use	A37		Use external termination plug
CAN_L_OUT	I/O	reserved for future use	B37		Use external termination plug
Shelf_EEPROM_WP	O	Shelf EEPROM write protection control	A38		Pull high on PMM, 10k to 3.3V
Power_KILL	I	Power Kill, short pin connect to GND via 10 Ohm on shelf	B38		short pin, for PMM hotswap power control
NC (clearance)	-	No copper finger	A39		NC
NC (clearance)	-	No copper finger	B39		NC
P48V_RTN (GND)	I	Ground	A40		
P48V_RTN (GND)	I	Ground	B40		
NC (clearance)	-	No copper finger	A41		
NC (clearance)	-	No copper finger	B41		
P48V_IN	I	48V Power In	A42		
P48V_IN	I	48V Power In	B42		

Table 3: PMM edge connector pinout assignment

4.4.2 Power supplies

The shelf provides 48V power to the PMM card through the edge connector. A “Power kill” short pin signal is also available on the PMM.

The PMM shall convert the 48V power input to the power rails needed for the MCU and other components. The PMM shall also provide 3.3V across the connector to the power shelf to power up the shelf manufacturing EEPROM.

One critical requirement for PMM power design is to avoid the arcing or damage during the PMM plug in and out when the shelf is powered up. The Power_Kill signal on the edge connector helps serve the purpose.

4.4.3 I2c for shelf manufacturing EEPROM

The PMM shall be able to read the shelf manufacturing data from the shelf EEPROM (address 0x50) through the i2c bus. The EEPROM is write-protected by default and can be enabled from the MCU if needed.

The same i2c bus can also be used to access the manufacturing EEPROM on the PMM itself. This EEPROM can be replaced by MCU interval flash but suggests keeping the option available on the PCB layout.

4.4.4 Power loss siren signal

The power loss siren signals from all modules on the shelf are wire ORed together and connected to pin 2 on #1 and #2 directly through the edge connector (no connecting to MCU).

The power loss siren signals from all shelves on the same Modbus daisy chain are wire ORed through the daisy chain.

The power loss signal is monitored by the rack management device on the ORV3 HPR rack.

4.4.5 Signals for Inter shelf control

ISHARE cable detection signals are routed to PMM for ISHARE cable detection.

CAN bus signals are routed to the PMM but not used in case it is needed to boost the CAN bus signal.

4.4.6 Modbus for module communication

The RS485 Modbus is used for data communication between the PMM and modules. This Modbus interface is called module Modbus hereafter.

Since the Modbus was used in ORV3 for module communication, using the Modbus for ORV3 HPR module communication can significantly reduce the module firmware changes. The module Modbus will operate at baud rate of 115200, with one termination resistor near the RS485 transceiver on PMM.

The module Modbus design follows the standard Modbus standard specification.

4.4.7 Shelf address for CAN bus addressing

Each BBU module on the CAN bus must have a unique CAN bus address. The signals shelf_addr_int_A<3:0> on the PMM edge connector will be used to generate the BBU CAN bus address.

The allocation of the 4-bit shelf address for CAN bus will be covered by “Modbus allocation table” in PMM software design specifications.

Since the CAN bus communication is running continuously, it might be necessary to keep the 4-bit address unchanged during PMM firmware upgrade or reboot. **So an I2C IO expander is suggested to control the shelf_addr_int_A<3:0>.**

4.4.8 PSU/BBU Module alerts and present signals

The PSU/BBU module alert and present signals are monitored by the MCU on the PMM. The status will become a part of shelf health status, available in registers to the rack management device. However, currently there is no plan for PMM to take any actions on these alerts.

4.4.9 Spare GPIOs

6 spare GPIOs are connected from MCU to the edge connector. They are not defined and not connected on the shelf side.

Suggest connecting the GPIOs to the test points on the shelf side.

4.5 Micro controller unit

The Micro Controller Unit (MCU) is the most important component on the PMM card. It will help the PMM to implement the following key tasks:

- Collects and consolidates the register data from the modules on the shelf periodically through the Modbus (as Modbus master).
- Collects and consolidates the status of modules within the shelf by monitoring some hardware signals.
- Collects manufacturing information about the shelf and the PMM itself through I2C etc.
- Prepare the module and shelf data for inquiry from the rack management device.
- Provide response to request from rack management device (as Modbus secondary).
- Allocate shelf address based on inputs from RJ45 port #1 (Pin status or serial communication).
- Share the shelf address to the next PMM/shelf through RJ45 port #2 (serial communication).
- Manage the module firmware upgrade. Provide firmware upgrade service to the rack management device.
- The MCU shall be able to keep and report the age of the cached data.

The selected MCU shall have enough processing power, memory, accessories and spare I/Os to support future growth without changing the MCU.

A couple of latched I/Os might be needed to keep some critical I/Os unchanged during the firmware upgrade or MCU reboot.

4.6 PMM Manufacturing EEPROM

PMM manufacturing information such as FBPN, vendor PN, Hardware and firmware version, etc shall be stored in the non-volatile memory, either in the MCU flash or EEPROM on the PMM. The contents shall NOT be changed during the firmware upgrade, unless the change is on purpose during the manufacturing process.

The information shall be available for inquiry from the Modbus.

5. PMM Modbus and CAN Bus Address Allocation

5.1 Inter PMM/shelf communication

Both Modbus and CAN bus communication require a unique address for each node on the daisy chain. Since all PMMs are manufactured the same, having the PMM with different pre-defined addresses is very hard to manage. The PMM design must be able to automatically allocate a unique PMM Modbus address or CAN bus address.

The MCU on the PMM can help to assign the PMM Modbus address if the MCU has the knowledge of the allocated Modbus addresses on the previous nodes. To solve the issue, a serial data communication is proposed to send the address information from one PMM to the next PMM on the daisy chain.

5.1.1 Serial data communication from one PMM to the next PMM on the daisy chain

The design is to use pin 7 on port #1 (or write as pin #1-7) to receive the data from the previous Modbus node (PMM) and use pin #2-7 to transmit the data to the next Modbus node (PMM).

Since the rack management device doesn't support this feature, the first PMM on the daisy won't be able to receive any serial data from pin #1-7. Since pin #1-6 is set to high by the rack management device, the PMM can use this to determine if the current PMM is the **first** PMM on the daisy chain.

All PMMs shall set the pin #2-6 to 0 (low) to indicate the availability of serial data communication to the next PMM.

The current view of the communication speed is ~9600 baud rate, one direction only.

5.2 Module Modbus address allocation

To simplify the on-shelf module Modbus communication, it is suggested to use the fixed Modbus address for all modules on the shelf based on the module's slot location. For example, use the 0b'100000zzz for modules on the shelf, where zzz is the slot address of the module. It can apply to both PSU and BBU shelf. So, the module Modbus address is 0x80 to 0x85 for a 6-slot shelf.

5.3 PMM Modbus address and CAN bus address allocation

The PMM Modbus address of all the PMMs/shelves on the same daisy chain must be unique to avoid address conflict.

With the serial communication capability described earlier, each PMM card can assign itself a Modbus address based on the information it receives from the previous shelf. The new information can be sent to the next shelf again. By doing so, all the PMMs on the daisy generate their own Modbus address one by one in the order of the locations on the daisy chain.

Same for the BBU CAN bus addressing, the PMM can assign the unique BBU CAN bus address based on the information received from the previous shelf on the daisy chain.

Please refer to PMM software/communication specifications for details of Modbus address and CAN bus allocation.

6. Communication with Rack Management Device

The PMM, as a Modbus secondary, provides Modbus service to the rack management device through the Modbus cable. The Modbus communication follows the standard Modbus specifications, including the register map definition and related Modbus functions.

Please refer to PMM software/communication specifications and register maps for details of communication between Rack management device, PMM and Modules.

6.1 Module Firmware upgrade support

Please refer to PMM software/communication specifications and register maps for details of module firmware upgrade.

6.2 PMM Firmware Upgrade

Please refer to PMM software/communication specifications and register maps for details of PMM firmware upgrade. Firmware upgrade shall not impact the operation of the power system.

7. Mechanical Requirements

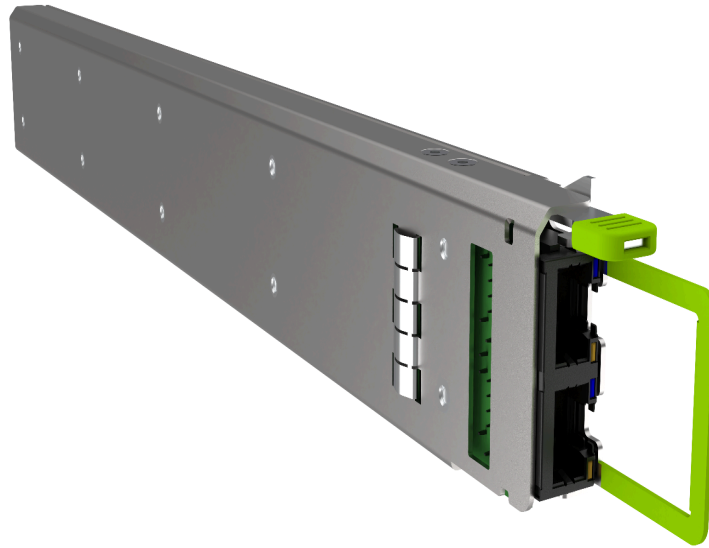


Figure 6: PMM Isometric View

7.1 PMM Physical Dimensions

The PMM is 15mm x 40.6mm x 350mm [Width x Height x Depth]

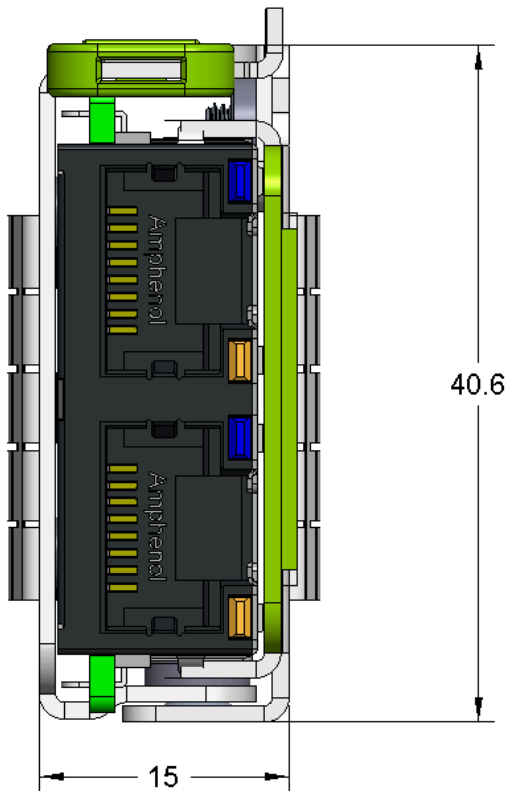


Figure 7: PMM Front View

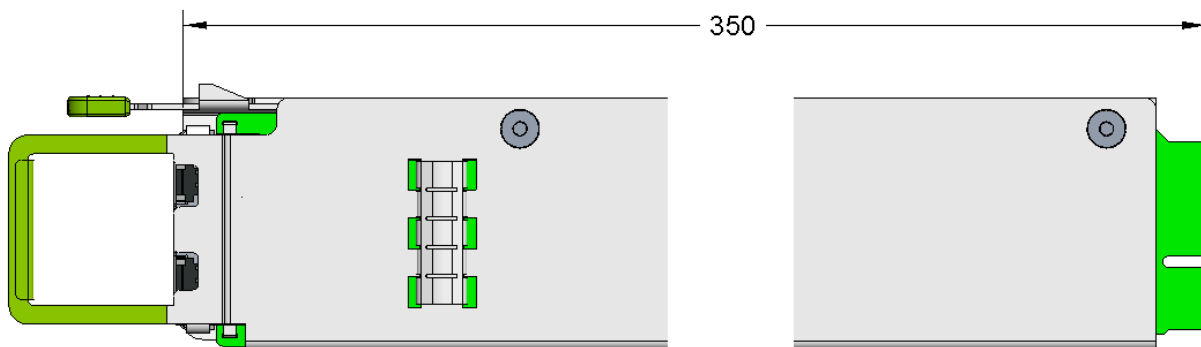


Figure 8: PMM Side View

7.2 Construction

The PMM base can be welded, riveted, or screwed together, consistent with meeting shock and vibration requirements. There shall be no sharp corners or edges.

The sheet metal material shall be pre-plated hot-dip zinc coated, with 0.8mm - 1.0mm of thickness.

7.3 Latch and Handle Requirements

A latch and handle are required for PMM removal and installation. The latch shall be attached in the location shown on the mechanical drawing and is to interface with a rectangular cutout on the HPRv2 PSU/BBU shelf chassis. The latch design may vary, but the finger interface of the latch must be Pantone 375C (green).

The handle is to be designed so it does not interfere with the removal and insertion of cables into the RJ45 connectors. Handle touch points are to be Pantone 375C (green).

7.4 EMI Gaskets

EMI gaskets are to be placed on the left & right and/or top & bottom sides of the PMM. Purpose of the gaskets is twofold:

1. PMM-to-shelf grounding
2. EMI containment

Gaskets are to be placed towards the front of the PMM so they make full contact with the walls of the shelf.

7.5 Materials

The sheet metal material shall be steel, pre-plated hot-dip zinc coated, with 0.8mm - 1.0mm of thickness unless otherwise specified. Any plastic material used should meet UL 94-V0 specifications. The system shall contain a minimum of 40% post-consumer-recycled (PCR) plastic content and 15% recycled steel content. However, functionality and reliability are paramount. 65% recycled PC/ABS and 20% recycled SGCC shall be qualified and used in applicable parts to meet the overall recycled content requirement.

7.5.1 PCR Targets & Implementation

- PCR content must be met in aggregate for polymers and steel.
- Functionality and reliability take priority; waivers may be considered for components where recycled content impacts performance.
- PCR requirements are mandatory at MP; strongly recommended to pilot in DVT for risk mitigation.

7.5.2 Supplier Qualification & Material Testing

- All new material suppliers or materials not previously validated must complete full qualification testing to confirm performance and compliance with recycled content requirements.
- Suppliers already validated for the specified recycled content do not require requalification.
- Required testing (as applicable):
 - Mechanical: Tensile, hardness, standoff mounting strength.
 - Corrosion/finish: Salt spray, silkscreen print adhesion.
 - Sheet metal feature limits: Max emboss height, min L-flange bend height, min closed hem length, min inner hole radius of closed hem, and other relevant features.
 - Key mechanical properties must be tested to relevant JIS standards (or equivalent), or any other applicable standards.

7.5.3 Documentation & Certification

- Required
 - Recycled Content Certificate from supplier.



Figure 9: Example of RC Certificate

- Material Certificate showing compliance with relevant standards (e.g., JIS G 3302:2019).
- Metal coating certificate (e.g., Z18–Z22).
- RoHS compliance certificate.
- BOM with detailed breakdown: material, PCR %, aggregate PCR % by weight for polymers and steel.
- Optional but Preferred
 - ISO 17065/14021 accreditation.
 - Recognition from LEED, EPA, or BREEAM.

7.6 Cover

The assembly shall have a sheet metal cover to protect board components from damage. The cover is held down by screws to the base and shall not interfere with any shelf features that are required for installation of the PMM.

7.7 RJ45 Connector

On the front side of the PMM, there is a single 2x RJ45 connector. This is a modular jack connector in a 1x2 configuration of RJ45s with LEDs. The MPN shall be Amphenol RJE72-2MF-0002-HF or equivalent. Note that this connector comes with spring fingers. These must contact the PMM chassis in order to provide sufficient EMI sealing.

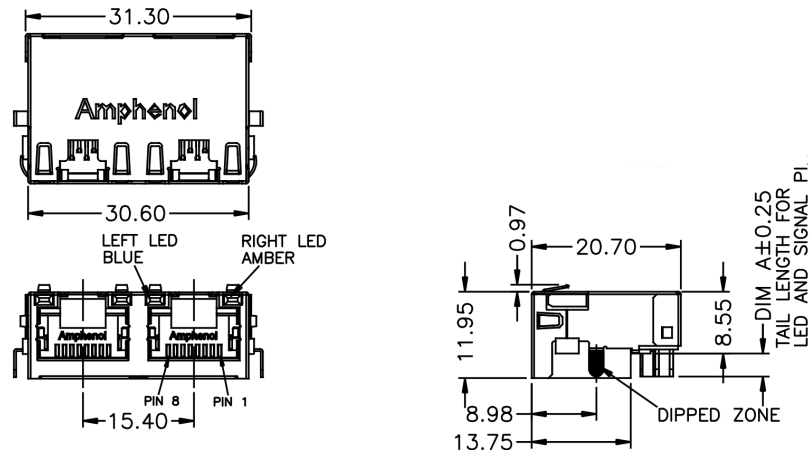


Figure 10: RJ45 Connector Drawing

7.7.1 RJ45 Port Numbering

When installed into the PMM chassis, the 1x2 RJ45 connector jack will be rotated 90 degrees. Port numbering shall follow the diagram shown below, with Port 1 being at the top and Port 2 being at the bottom.

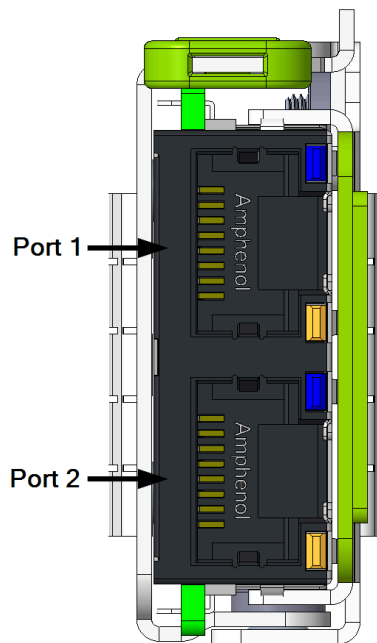


Figure 11: RJ45 Port Numbering

7.8 LED on the Front panel

Due to the limited size of the PMM front panel, the PMM will use LEDs on the RJ45 ports as its indicators. Blue LEDs on the left side of RJ45 ports will be controlled at the same time as a single blue LED indicator.

Amber LEDs on the right side of RJ45 ports will be controlled at the same time as a single amber LED indicator.

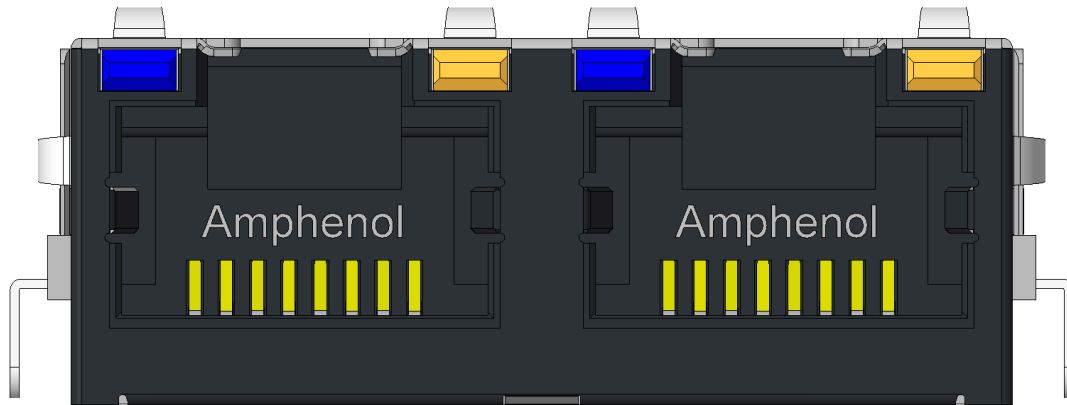


Figure 12: RJ45 LEDs

The behavior of the blue and amber LED is described below.

States	Blue LED 	Amber LED 	Meaning
1	Off	Off	No power to PMM or PMM Power Failure
2	Solid on	Off	PMM operation normal
3	Solid on	x	PMM ready (not including wiring checking)
4	Blinking	x	PMM firmware upgrade
5	Blinking or solid on	Blink (0.25s/0.25s)	ISHARE cable not detected (Does not apply to the first shelf) (priority 3, see note)
6	Solid on	Blink (0.5s/0.5s)	No valid PMM Modbus address. (Serial link doesn't receive valid data and A2=0) (priority 2)
7	Solid on	Solid on	PMM in unknown shelf type (priority 1)
8	Off	Solid on	PMM permanent failure (but power is OK to drive the LED)(priority 0,highest)

Table 4: LED indication

Note:

When there are two or more failures happening at the same time, use the LED to indicate the failure with the highest priority. Priority 0 represents the highest priority.

7.9 Blind Mate Connector

The PCB (card edge) of the PMM module blind mates into a right angle connector on the shelf side. The location of this connector is fixed, but the connector itself may either be mounted on a PCB or a panel-mounted cable. There are no specific requirements for the interface of this blind mate connector to the main shelf PCB. Please refer to the “*Open Rack V3 HPR V2 72kW Power Shelf*” [1] or “*Open Rack V3 HPR V2 72kW BBU Shelf*” [2] specifications for more details on this.

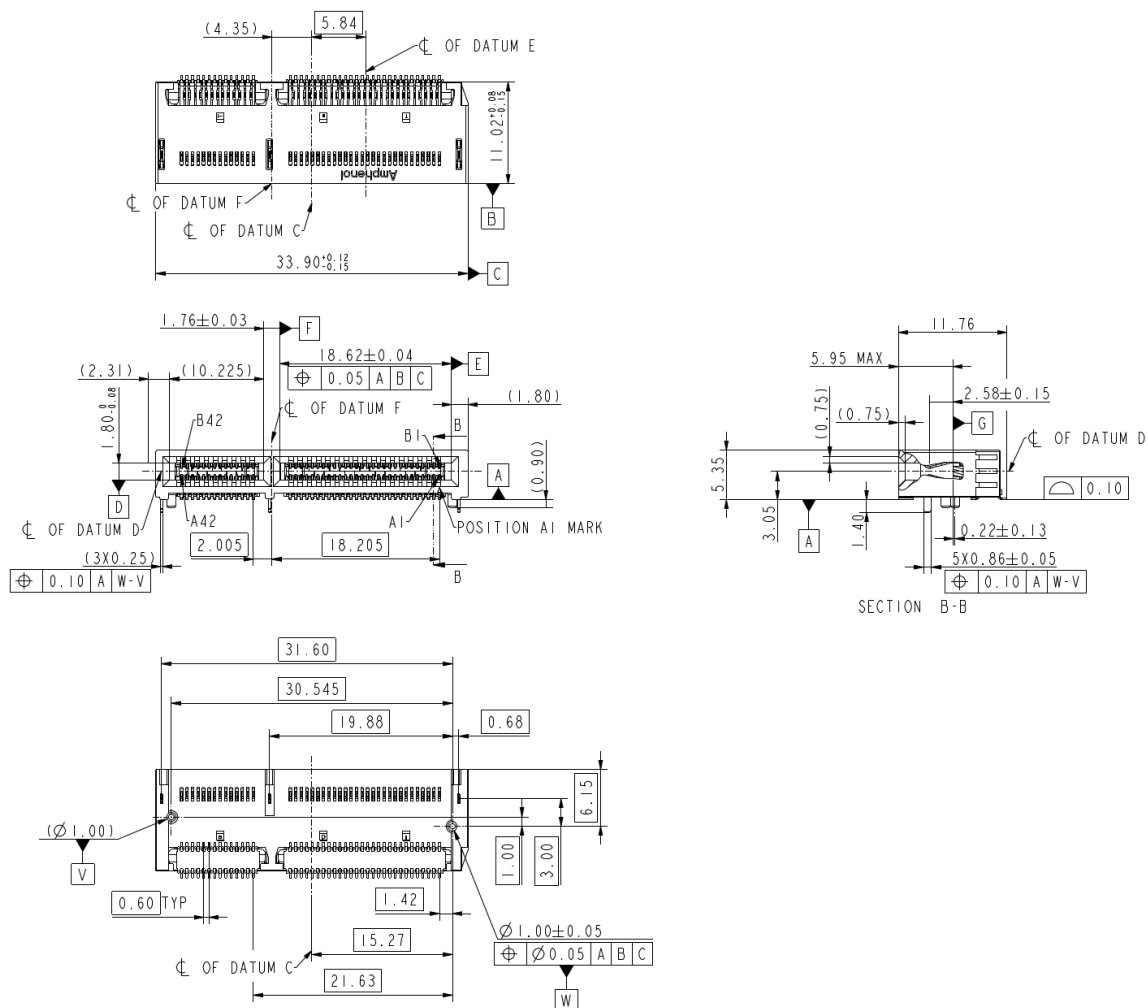
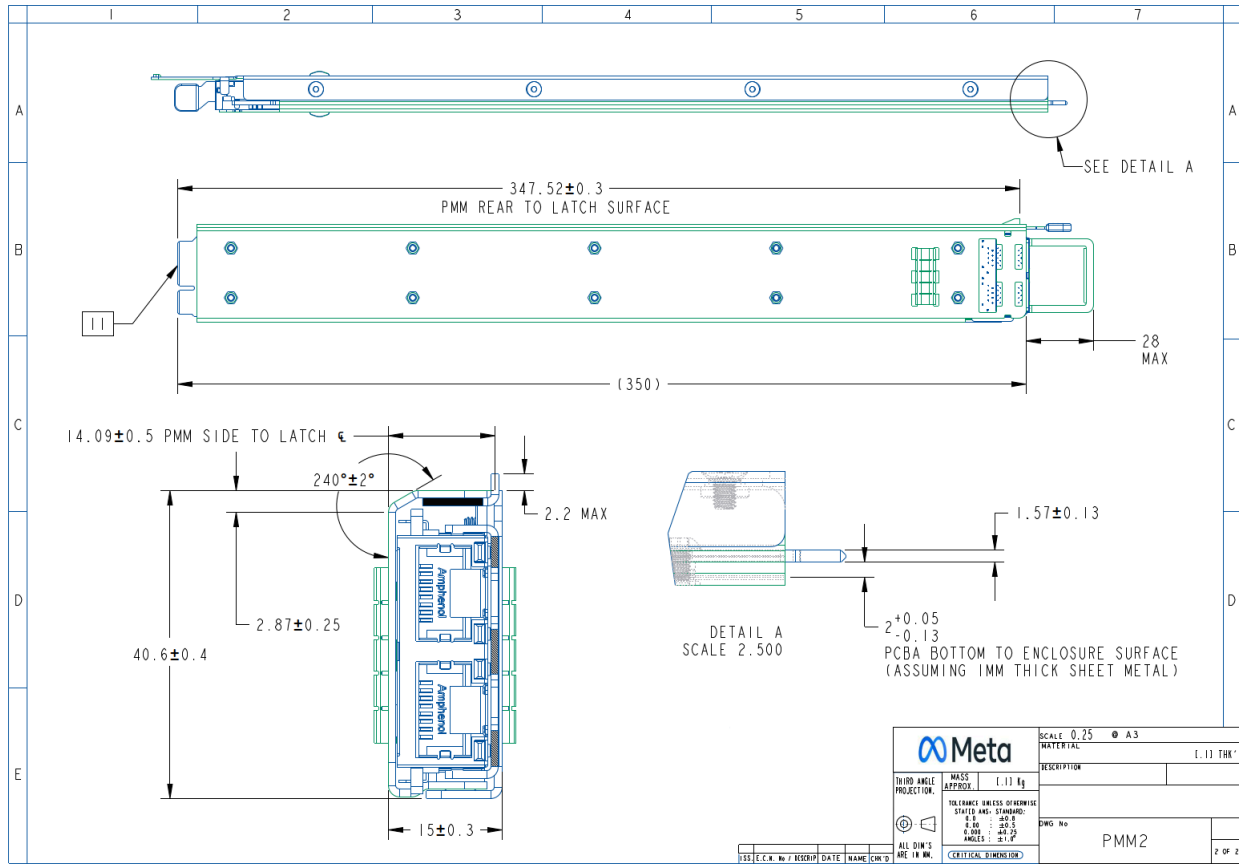


Figure 13: PMM Connector Drawing

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8. Regulatory Compliance/Safety Requirements

The shelf power monitoring module (PMM) shall be designed for compliance to allow worldwide deployment. Additionally, the manufacturer is fully responsible for:

- Ensuring the complete compliance of the power supply shelf in the environment it is intended to function (as described by the Rack Spec)
- Maintaining and updating the power supply shelf safety reports to current requirements and all new released requirements.
- All design and recertification costs required to update the power supply to meet new requirements
- Meeting EMC requirements
- Meeting Safety requirements

The manufacturer is responsible for obtaining the safety certifications specified in the following sections.

8.1 Safety Standards

Power/Battery Shelf Management Controller (PMM) shall be tested together in the OCP ORV3 HPR V2 power shelf and shall comply with all safety requirements specified in “Open Rack V3 HPR V2 72kW Power Shelf” [1] and “Open Rack V3 HPR V2 72kW BBU Shelf” [2] specifications.

Any component or signal that controls charging and discharging battery shall be evaluated under single fault condition per UL62368-1 Annex M.

8.2 EMC Requirements

Power/Battery Shelf Management Controller (PMM) shall be tested together in the OCP ORV3 HPR V2 power shelf and shall comply with all EMC requirements specified in “*Open Rack V3 HPR V2 72kW Power Shelf*” [1] document when PMM board runs in a normal operation mode.

8.3 Environmental Compliance

Manufacturer of PMM shall provide full material disclosure, and technical documentation to demonstrate compliance to environmental compliance requirements such as RoHS, REACH, WEEE etc, depending on the end user’s goals and business needs.

8.4 Documentation

The PMM shall be covered under all documents specified in “*Open Rack V3 HPR V2 72kW Power Shelf*” [1] document.

9. Serviceability

9.1 Required collateral

Shall include (at minimum) schematics, CAD, block diagrams, I2C diagram, switch and jumper settings, service/repair/assembly/disassembly instructions (if applicable), LED definitions.

9.2 FRU information

FRU labels will include part number and serial number. FRU EEPROM shall be per IPMI spec.

9.3 PCB information

PCBs shall have silkscreen markings for jumpers, LEDs, and FRUs.

9.4 Port Symbols

The Management Ethernet connectors shall have the text “MGMT” printed on or the metal stamped “MGMT” near the dedicated network port and include appropriate pictogram, and Sideband RJ45 shall have the text “SB” printed on or the metal stamped “SB” near the dedicated network port and include appropriate pictogram.

9.5 Cold Aisle Repairs

PMM shall be serviced from the Cold Aisle.

9.6 Programmable updates

Non-disruptive (no reboot required) update capability is required for all programmable devices. Utilities shall be provided by the Design Partner with the capability to up and down-rev each article of firmware in the equipment.

9.7 Additional Service Requirements

Additional Service requirements may be added based on Safety and Data Center requirements.

10. Thermal

10.1 Environmental (operational)

- Cold aisle (inlet) temperature: -5°C to 45°C
- Relative humidity: 10% to 90%, non-condensing
- Altitude: up to 3050m (10,000ft)
- Acoustic noise: If active cooling is required, target sound pressure should not exceed 85dBA when fan modules are running at full speed and operating within the defined environmental envelope.

10.2 Thermal Requirements

- Operational
 - Thermal margin: Reserving adequate margins on components is critical. These margins should be defined with respect to de-rated values, as appropriate. Following are the requirements.
 - Component thermal margin of $\geq 7\%$ or $\geq 5^\circ\text{C}$ up to 30°C inlet/ambient and 3050m (10,000ft) above sea-level. Target whichever value is larger.
 - Component thermal margin of $\geq 4\%$ or $\geq 3^\circ\text{C}$ at greater than 30°C inlet/ambient and up to 3050m (10,000ft) above sea-level. Target whichever value is larger.
 - Margin to de-rated temperatures should account for associated differences in reading and measurement location. Impact on reliability should also be considered when determining required margin.
- Thermal kit
 - Sensor accuracy: For discrete and critical sensors (such as ambient temperature), an accuracy of $\pm 2^\circ\text{C}$ is required ($\pm 1^\circ\text{C}$ is preferred). If a component does not have an integrated temperature sensor, and uses a proxy, need to target an accuracy $\pm 5^\circ\text{C}$ ($\pm 2^\circ\text{C}$ is preferred). If this component is temperature sensitive, thermal margin requirements defined above should account for sensor inaccuracy.
 - System fan (if required): Should be sized to support operation across environmental and loading envelopes, with an adequate operating range (speed) to achieve requirements outlined in this document. The fan should also have adequate overhead to accommodate back-pressure resulting from shelf design, rack-level accessories and data center operation. In general, head room to overcome a back-pressure of ≥ 0.3 inches of water is highly recommended. This does not take into account impedance created at the front of the shelf to entrain airflow in the PMM (if required). In addition, to ensure thermally-efficient operation, airflow greater than 5CFM should be avoided.

- Surface temperature: To make the PMM safe for handling in-operation, accessible surfaces should not exceed a temperature of 70°C.

11. Vibration and Shock (Non-Packaged)

The PMM/PMC Module shall meet vibration and shock test per EN 60068-2-6 and 60068-2-27 respectively and is to be performed at system shelf level (i.e. power shelf installed with PSUs and PMM/PMC module).

Please refer to the Vibration & Shock sections of “*Open Rack V3 HPR V2 72kW Power Shelf*” [1] or “*Open Rack V3 HPR V2 72kW BBU Shelf*” [2] specifications.

12. Package Vibration, Drop and Compression

PMM/PMC Modules units in their package shall meet the following requirements:

Package Vibration	1.146 G _{rms} , 2-200-2 Hz, all three axes, Random Vibe	ISTA 3E 06-06
Package Drop	8-inch drop	ISTA 3E 06-06
Package Compression	Maximum compression loading on a bulk pack	ASTM D 642-94

13. Compliance with OCP Tenets

13.1 Openness

This specification publishes the complete set of requirements needed for any qualified third party to design, build, and service a compliant PMM, including the electrical interfaces, edge-connector and RJ45 pinouts, Modbus and CAN bus addressing scheme, microcontroller and EEPROM requirements, and firmware-upgrade behavior.

13.2 Efficiency

By reusing the existing ORv3 communication interface, the PMM minimizes the changes required to the PSU and BBU register maps and firmware, reducing development effort and easing adoption across the ecosystem. A single PMM per shelf, combined with automatic address allocation across a multi-shelf daisy chain, simplifies rack-level monitoring and reduces cabling and configuration overhead.

13.3 Impact

The PMM is the standardized monitoring building block of the Open Rack V3 High Power Rack (HPR) ecosystem, providing the rack management device with consolidated power and health telemetry from every PSU and BBU shelf.

13.4 Scale

The PMM supports multi-shelf Modbus daisy chains with automatic allocation of unique Modbus and CAN bus addresses across mixed PSU and BBU shelves, enabling monitoring to scale from a single rack to large hyperscale deployments without manual configuration. It is hot-swappable and supports online fail-safe firmware upgrades, for both itself and the modules, with no power interruption at the rack level, supporting serviceability at scale.

13.5 Sustainability

The specification defines mandatory post-consumer recycled (PCR) content targets, met in aggregate by weight for polymers and steel, together with supplier qualification, recycled-content certification, and a BOM-level PCR breakdown to verify them. The PMM conforms to applicable environmental regulations, including RoHS and REACH, supported by full material disclosure.

14. References

Refer to the latest available revision of the following specifications:

- [1] *Open Rack V3 HPR V2 72kW Power Shelf* (in progress as of June 2026)
- [2] *Open Rack V3 HPR V2 72kW BBU Shelf* (in progress as of June 2026)
- [3] Modbus Application Protocol V1.1b3